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Copper-catalyzed aerobic oxidation of primary alcohols to carboxylic acids†

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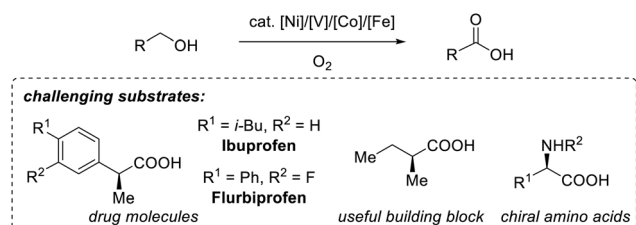
Here, the first copper-catalyzed aerobic oxidation of primary alcohols to carboxylic acids with TEMPO and KHSO₄ as the co-catalysts has been developed. The reaction exhibits excellent substrate scope and functional group compatibility under mild conditions. Even the very sensitive chiral alcohols, chiral amino alcohols, and alcohol-containing steroid skeletons may be oxidized to afford the corresponding carboxylic acids or lactones without racemization.

Carboxylic acids are prevalent in natural products, agrochemicals and drug molecules, such as ibuprofen, flurbiprofen, naproxen *etc.*¹ Furthermore, carboxylic acids are widely featured in the synthesis of other chemicals of academic and/or industrial interest.² One of the most common protocols for synthesizing carboxylic acids is the oxidation of primary alcohols.^{3–5} However, at least stoichiometric amounts of toxic and expensive oxidants are required in the traditional approaches, which is not cost-effective and environmentally friendly due to the high cost of such oxidants and the corresponding derived hazardous wastes.^{4,6,7} In this regard, the development of catalytic strategies using the environmentally benign and inexpensive molecular oxygen as the oxidant has been prominent.^{5,8–10} Some such methods with metal catalysts have been reported.^{11,12} However, the substrate scope is still very limited: the challenging substrates are the chiral alcohols including drug molecules, (*S*)-2-methylbutanol,¹³ and chiral amino alcohols¹⁴ due to very sensitive nature of the corresponding chiral acids with an α -chirality under the employed reaction parameters (Scheme 1a). Thus, we turned our attention to copper catalysis. The oxidation of primary alcohols to carboxylic acids with very high loadings of terminal oxidants,

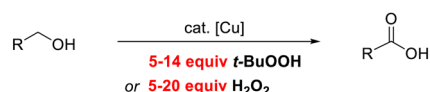
such as, TBHP (5–14 equiv)) or H₂O₂ (5–20 equiv), has been reported (Scheme 1b).¹⁵ So far, there are no reports on the copper-catalyzed aerobic oxidation of primary alcohols to carboxylic acids.^{16,17} Herein, we present the first copper-catalyzed aerobic oxidation of primary alcohols to carboxylic acids under very mild conditions with a remarkable functional group compatibility under the assistance of a catalytic amount each of TEMPO and KHSO₄ (Scheme 1c).

We commenced our investigations with the evaluation of different catalysts to enable the direct aerobic oxidation of alcohols to carboxylic acids by using cetyl alcohol (**1a**) as the model substrate (Table 1). Aldehyde **2a'** was formed in a 61% yield with 21% of recovery of **1a** when the reaction was

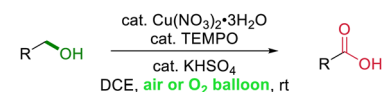
(a) The earth-abundant metals-catalyzed aerobic oxidation of primary alcohols to carboxylic acids



(b) Copper-catalyzed oxidation of primary alcohols to carboxylic acids with high-loadings of oxidants



(c) Copper-catalyzed aerobic oxidation of alcohols to carboxylic acids (this work)



- ◆ readily available catalysts
- ◆ broad substrate scope
- ◆ very mild conditions
- ◆ chiral acid syntheses

Scheme 1 (a) The known earth-abundant metal-catalyzed aerobic oxidation of primary alcohols to carboxylic acids; (b) and (c) copper-catalyzed oxidation of primary alcohols to carboxylic acids.

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Table 1 Screening of an extra catalyst for the aerobic oxidation of **1a**^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH} \xrightarrow[\text{DCE, O}_2 \text{ (balloon), rt, 36 h}]{\text{Cu(NO}_3)_2 \cdot 3\text{H}_2\text{O (10 mol\%)} \\ \text{TEMPO (10 mol\%)} \\ \text{additive (10 mol\%)}} n\text{-C}_{15}\text{H}_{31}\text{CHO} + n\text{-C}_{15}\text{H}_{31}\text{COOH}$				
Entry	Additive	Recovery of 1a (%)	NMR yield of 2a' (%)	NMR yield of 2a (%)
1	None	21	61	0
2	KHCO ₃	95	5	0
3	K ₂ CO ₃	93	5	0
4	K ₃ PO ₄	94	6	0
5	K ₂ HPO ₄	95	6	0
6	KCl	0	18	76
7	KBr	0	61	30
8	KI	67	31	0
9	K ₂ SO ₄	48	35	0
10	KH ₂ PO ₄	0	12	87
11	NaH ₂ PO ₄	0	0	98
12	NaHSO ₄	0	0	99
13	H ₂ O	0	0	100
14	KHSO ₄	0	6	90
15	AlCl ₃	0	4	92
16 ^b	ZnCl ₂	0	0	0
17 ^c	KHSO ₄	100	0	0
		0	10	88

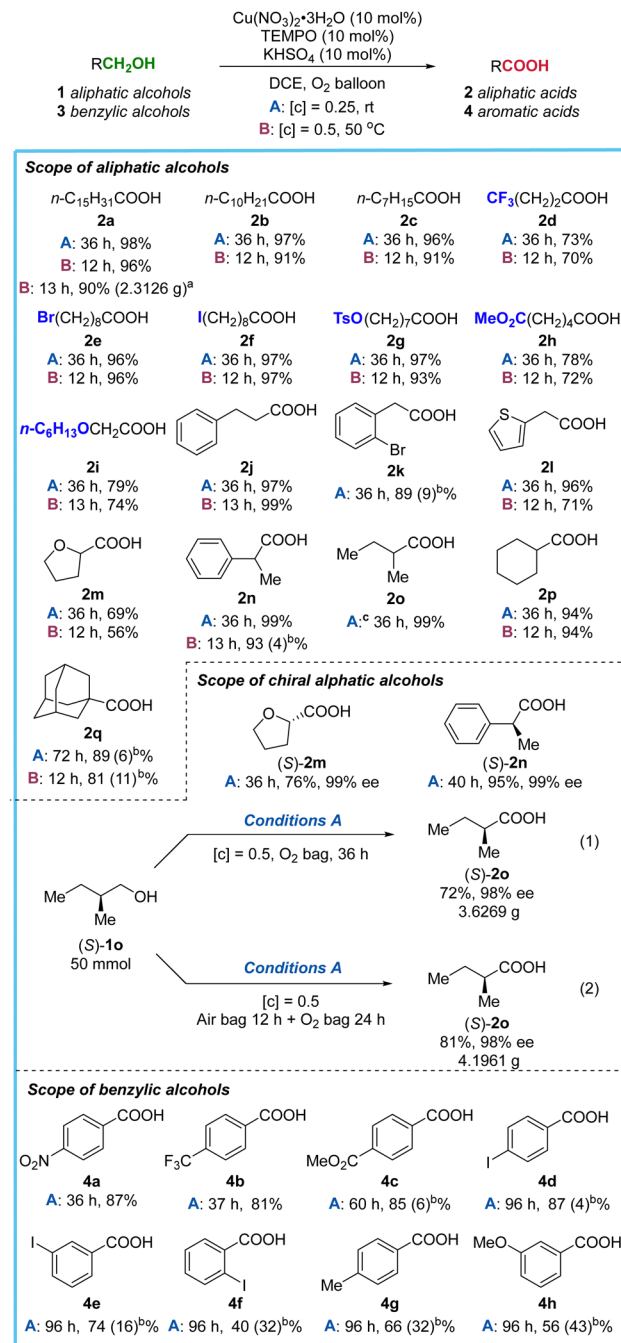
^a The reaction was conducted with 1.0 mmol of **1a**, 10 mol% each of Cu(NO₃)₂·3H₂O, TEMPO, and additive in 4 mL of DCE at rt for 36 h with an O₂ balloon. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

^b Without Cu(NO₃)₂·3H₂O. ^c The reaction was conducted for 72 h with an air balloon.

conducted with 10 mol% of Cu(NO₃)₂·3H₂O and TEMPO (entry 1).^{17c} KHCO₃, K₂CO₃, K₃PO₄, and K₂HPO₄, were first tested with considerable recovery of starting alcohol **1a** (entries 2–5). Potassium halides have also been screened, and among them, KCl is better than KBr and KI albeit all with poor results (entries 6–8). Further screening led to the observation that the addition of KHSO₄ yielded carboxylic acid **2a** in a quantitative yield (entry 13). A range of Lewis acids and solvents were also tested (Tables S1 and S2, ESI[†]), more than 90% of the corresponding acid **2a** was formed in the presence of AlCl₃ or ZnCl₂ (entries 14 and 15). Control experiment indicated that no reaction occurred in the absence of copper (entry 16) and a much lower efficiency was observed when the reaction was conducted in an air atmosphere (entry 17).

With these optimal reaction conditions in hand, the substrate scope was evaluated (Scheme 2). In addition to non-functionalized aliphatic acids **2a–c**, the reaction may tolerate various functional groups, such as halogens, esters, ethers, and aryl and heteroaryl groups, affording the corresponding carboxylic acids **2d–m** in up to 99% yields. Besides, the sterically more hindered 2-phenylpropan-1-ol, 2-methylbutan-1-ol, cyclohexyl methanol, and even 1-adamantanemethanol could be converted to the corresponding acids **2n–q**. Notably, chiral acids (*S*)-**2m** and (*S*)-**2n** could be formed efficiently without erosion of enantiomeric excess as compared with the starting chiral alcohols.

To demonstrate the practicality of this protocol, some gram-scale reactions were executed. When 10 mmol of cetyl alcohol **1a** was submitted to the standard conditions at 50 °C, 2.3 g



Scheme 2 Substrate scope of aliphatic alcohols and benzylic alcohols. Conditions A: The reaction was conducted with a 1.0 mmol scale of alcohol in 4 mL of DCE at rt with an O₂ balloon. Conditions B: The reaction was conducted with a 1.0 mmol scale of alcohol in 2 mL of DCE at 50 °C with an O₂ balloon. ^a The reaction was conducted with 10 mmol scale of alcohol in 20 mL of DCE at 50 °C with an O₂ balloon. ^b NMR yields of the aldehydes in parentheses were determined by ¹H NMR analysis using dibromomethane as the internal standard. ^c NMR yields were determined by ¹H NMR analysis using dibromomethane as the internal standard.

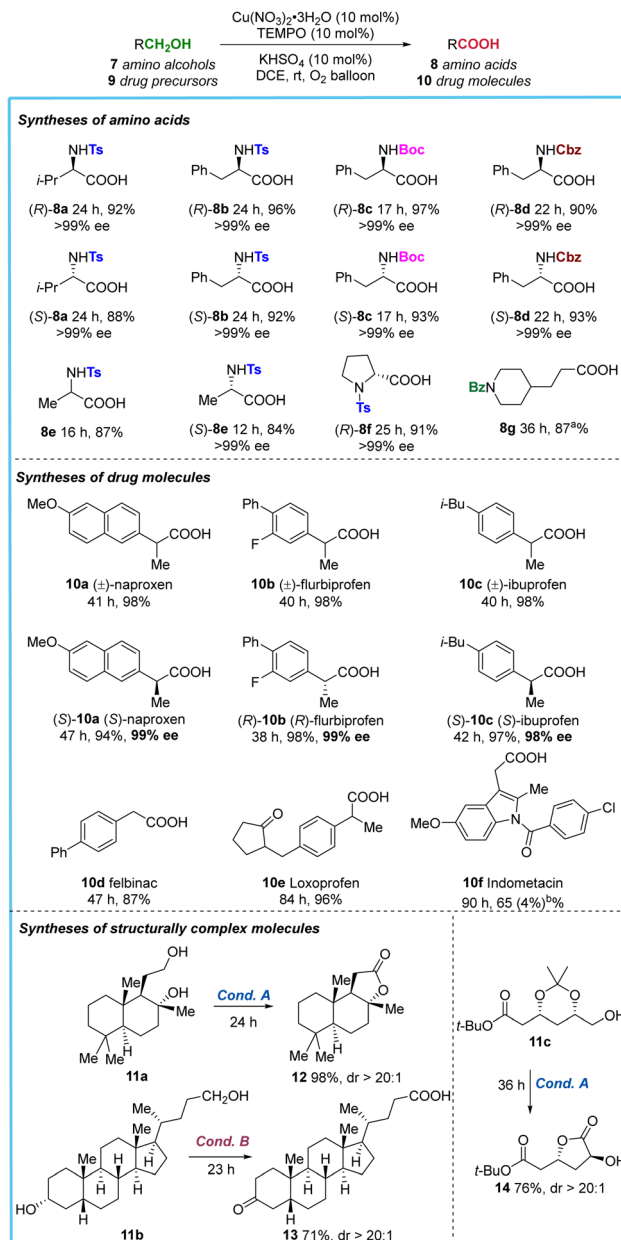
(90% yield) of **2a** could be afforded *via* simple recrystallization. (*S*)-2-methylbutanoic acid is an important chiral building block and very expensive (¥ 1393 g^{−1} from Aldrich). Jones oxidation of (*S*)-2-methylbutanol, which utilized the highly toxic chromium

compounds, is one of the traditional approaches to afford the chiral acid.¹³ Benefiting from the green and mild protocol developed here, a 50 mmol scale reaction of (*S*)-2-methylbutanol (*S*)-**1o** readily available from Aldrich (¥ 869/100 mL) could smoothly provide 3.6 g (72% yield) of (*S*)-**2o** *via* a simple distillation (eqn (1)). Also, the same reaction could be performed with a yield of 81% by using an air bag for 12 h first, which was followed by the supplementary supply of oxygen from a pure O₂ bag (eqn (2)).

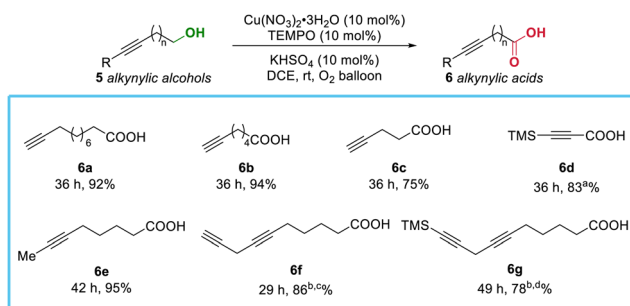
Next, benzylic alcohols were tested. The reaction is amenable to electron-withdrawing groups substituted at the *para*-position on the phenyl ring. Various functional groups, including nitro (**4a**), trifluoromethyl (**4b**), ester (**4c**), and halogen (**4d**) are well tolerated. We also found that the reactivity of substrates bearing the iodo group at the *o*-, *m*-, and *p*-positions increased as the steric hindrance decreased (**4d-f**). However, a mixture of the aldehydes and acids was obtained when the phenyl rings were substituted with electron-donating groups, such as methyl (**4g**) or methoxy (**4h**) groups.

We also explored the scope of synthetically attractive alkylnic alcohols (Scheme 3). The alcohols bearing a terminal C≡C bond unit *via* different lengths of carbon chains could be oxidized to the corresponding acids in good to excellent yields (**6a-c**). TMS-substituted propargyl alcohol could also be converted to the corresponding acid **6d** with a slightly higher loading of TEMPO. The methyl- or propargyl-substituted hept-6-yl-1-ol afforded the corresponding acids **6e-g** *via* varying the catalyst loading.

Oxidation of alcohols with a chiral center at the α-position is very challenging due to the possible racemization of the corresponding rather sensitive carboxylic acid products: (1) amino alcohols are very attractive substrates for aerobic oxidation.¹⁴ The current protocol has been applied successfully to provide the corresponding amino acids **8a-g** smoothly in excellent yields (Scheme 4). (2) Furthermore, this methodology could be applied for the synthesis of racemic drug molecules, such as naproxen **10a**, flurbiprofen **10b**, ibuprofen **10c**, felbinac **10d**, loxoprofen **10e**, and indomethacin **10f**, from the corresponding



Scheme 4 Syntheses of amino acids, drug molecules, and structurally complex molecules. The reaction was conducted with a 1.0 mmol scale of alcohol in 4 mL of DCE at rt with an O₂ balloon. ^aThe reaction was conducted with 0.5 mmol scale of alcohol in 2 mL of DCE at rt with an O₂ balloon. ^bNMR yield of the aldehyde in parentheses was determined by ¹H NMR analysis using dibromomethane as the internal standard. For conditions A and conditions B, see footnotes of Scheme 2.



Scheme 3 Substrate scope of alkylnic alcohols. The reaction was conducted with 1.0 mmol scale of alcohol in 4 mL of DCE at rt with an O₂ balloon. ^a10 mol% Cu(NO₃)₂·3H₂O, 12 mol% TEMPO and 10 mol% KHSO₄ were used. ^bThe reaction was conducted with 0.5 mmol scale of alcohol in 2 mL of DCE at rt with an O₂ balloon. ^c20 mol% Cu(NO₃)₂·3H₂O, 25 mol% TEMPO and 20 mol% KHSO₄ were used. ^d20 mol% Cu(NO₃)₂·3H₂O, 30 mol% TEMPO and 20 mol% KHSO₄ were used.

alcohols in good to excellent yields. In addition, the corresponding optically active drug molecules, such as (*S*)-naproxen, (*R*)-flurbiprofen, and (*S*)-ibuprofen were also prepared without racemization. Interestingly, sclareolide **12** could be efficiently synthesized *via* the sequential oxidation and lactonization of the tertiary alcohol unit in the molecule. Alcohol **11b** containing a steroid skeleton could be oxidized to **13** in 71% yield with the secondary alcohol being converted to the ketone as well. Of note, chiral substrate **11c** with a ketal structure unit could also

undergo a highly efficient sequential aerobic oxidation, deacetalization, and lactonization to produce lactone **14** with a free hydroxyl group under the standard conditions.

In conclusion, we have developed the first copper nitrate-catalyzed aerobic oxidation of alcohols to carboxylic acids with the help of TEMPO and KHSO_4 . The reaction is applicable to a very wide range of alcohols, such as aliphatic alcohols, benzylic alcohols, alkynyl alcohols, amino alcohols, drug precursors, and structurally complex molecules, providing the corresponding carboxylic acids in good to excellent yields. Benefiting from the mild conditions, optically active acids could be prepared from chiral amino alcohols, chiral drug precursors, and structurally complex chiral molecules efficiently without racemization. Furthermore, the oxidation reaction could be readily conducted on a practical scale (50 mmol) for the synthesis of not readily available (*S*)-2-methylbutanoic acid with a similar efficiency. Further studies including the roles of Cu and KHSO_4 are being actively pursued in our laboratory.

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Conflicts of interest

There are no conflicts to declare.

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